Failure mode measured on the VCS	VCS measured evidence proven in this benchmark		Neural-network analogue documented, not proven here	Status here → there
1 **Gradient vanishes on discrete / index outputs**	**0.222** gradient bucket F (position) single record	The NAIVE position gradient is provably zero at a strobe-timed argmax (§3.2); the whole gradient bucket stays low, 0.222, on the position regime.	Shattered / saturated gradients; non-differentiable argmax & index readouts.	proven here analogue there
2 **Saliency is model-invariant (sanity-check fails)**	**0.000** sanity-check margin single record	Program-randomization: saliency cosine = 1.000 while 114/128 RAM bytes changed -> sanity check FAILS.	Saliency unchanged under model- & label-randomization; behaves like an edge detector.	proven here analogue there
3 **Correlation ≠ causation (Granger & tuning trap)**	**0.157** Granger F 95% CI [0.11, 0.20]	Granger-edge faithfulness = 0.157 (false edges vs the true data-flow); game-variable tuning = 0.207.	Spurious Granger edges; 'present ≠ used' tuning & attention read as explanation.	proven here analogue there
4 **Single-unit ablation has an interaction blind-spot**	**0.213** connectome F 95% CI [0.14, 0.29]	Single-unit lesion is faithful (0.964) yet misses the wiring: connectome recovered at only 0.213.	Single-unit ablation misleading; computation is distributed across many units.	proven here analogue there
5 **SAE/probe: present-not-used**	**0.211** selectivity F 95% CI [0.14, 0.30]	SAEs match a variable, yet score 0.147 on causal use; probes decode at 0.950 acc but 79 cells are not causal.	Control-task selectivity gap; superposition / polysemantic neurons.	proven here analogue there
6 **IG/EG baseline sensitivity**	**0.186** EG F (all-regime) 95% CI [0.13, 0.25]	IG magnitude is baseline-sensitive (up to 1.000 across games); a content-byte baseline kills IG; EG lands at 0.186.	Attribution depends on the (arbitrary) baseline choice; no canonical reference input.	proven here analogue there